Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) Give Strassen's matrix multiplication algorithm and derive the time complexity.
b) Illustrate with an example the working of quick sort. [7+8]

2.a) Explain various asymptotic notations for algorithm analysis.
b) Derive the time complexity of binary search. [7+8]

3. Give the solution to the 8-queens problem using backtracking. [15]

4. Explain the solution to the graph coloring problem using backtracking. [15]

5. Explain the Single source shortest path problem with an example. [15]

6. What is the need for generating a spanning tree? Explain an algorithm for generating spanning tree. [15]

7. Discuss the time and space complexity of Dynamic Programming traveling sales person algorithm. [15]

8. Explain the LC branch and bound algorithm. [15]

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